

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/B00T or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull-downs to GND (PROJECT DESIGN CHOICE); reset mode M1/M0=00. AUX is input-only, no pull-down.

OLED: GPIO21/SDA and GPIO22/SCL are signal-only reservations; OLED VCC and I2C pull-ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 -> SN74AHCT1G125DBVR (OE=GND, V5) -> WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE; TPD45311 CC protection is not instantiated: its mandatory 2.7-4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test-point footprints/locations are deferred.

The schematic diagram illustrates the Stage 5 modular carrier for the ESP32-E220 LoRa Receiver. It features several key components and connections:

- DevKit Interfaces:** J1 and J2 represent DevKit USB-C ports. J1 is labeled "DEVKIT_LEFT (USB-C toward antenna)" and J2 is labeled "DEVKIT_RIGHT (USB-C toward antenna)". Both have pin headers with signals like GND, 5V_SYS, DP_NC, DM_NC, and various control lines.
- E220-T22D Socket:** J3 is a universal E220-T22D interface socket, noted as fitting either 400T22D or 900T22D modules. Its pins include E220_M0, E220_M1, E220_RXD, E220_TXD, E220_AUX, 5V_SYS, and GND.
- Power Management:** The design includes a 3V3 buck converter (U1), a USB-to-UART bridge (U2), and a USB-to-UART bridge (U3). Power rails like 5V_SYS, GND, and 3V3 are distributed throughout the board.
- Control and Data Lines:** Various control lines such as E220_TXD, E220_RXD, E220_AUX, and E220_M0/E220_M1 are shown connecting different functional blocks.
- Test Points:** Several test points (TP1 through TP7) are indicated for debugging and testing, corresponding to specific signals like 5V_SYS, E220_VCC, E220_TXD, M1, AUX, E220_RXD, and E220_TXD.
- Component Footprints:** Numerous component footprints are shown, including resistors (R1-R10), capacitors (C1-C10), and integrated circuits (ICs) like the ESP32 (U1), TPD1E10B06DPYR (U2), TPD45311 (U3), and various logic chips (U4, U5, U6, U7, U8, U9, U10, U11, U12, U13, U14, U15, U16, U17, U18, U19, U20, U21, U22, U23, U24, U25, U26, U27, U28, U29, U30, U31, U32, U33, U34, U35, U36, U37, U38, U39, U40, U41, U42, U43, U44, U45, U46, U47, U48, U49, U50, U51, U52, U53, U54, U55, U56, U57, U58, U59, U60, U61, U62, U63, U64, U65, U66, U67, U68, U69, U70, U71, U72, U73, U74, U75, U76, U77, U78, U79, U80, U81, U82, U83, U84, U85, U86, U87, U88, U89, U90, U91, U92, U93, U94, U95, U96, U97, U98, U99, U100).

The diagram also includes a title block with the following information:

- Sheet: /
- File: esp32-e220.kicad_sch
- Title: ESP32 E220 LoRa Receiver – Stage 5 modular carrier
- Size: A4
- Date:
- KiCad E.D.A. 10.0.5
- Rev: Id: 1/1

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/B00T or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull-downs to GND (PROJECT DESIGN CHOICE); reset mode M1/M0=00. AUX is input-only, no pull-down.

OLED: GPIO21/SDA and GPIO22/SCL are signal-only reservations; OLED VCC and I2C pull-ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 -> SN74AHCT1G125DBVR (OE=GND, V5) -> WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE; TPD45311 CC protection is not instantiated: its mandatory 2.7-4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test-point footprints/locations are deferred.

The diagram shows a KICAD schematic for a PCB titled "ESP32-e220.kicad_sch". It features several components and their connections:

- Connectors:** J1 and J2 are DevKit USB-C sockets. J3 is an E220-T22D socket. J4 is a USB4105-GF-A connector.
- Microcontrollers/Processors:** U1 is an ESP32 module. U2 is a GRM188R61A106MA0R microcontroller. U3 is a SN74AHCT1G125DBVR buffer.
- Resistors:** R8, R9, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C48, C49, C50, C51, C52, C53, C54, C55, C56, C57, C58, C59, C60, C61, C62, C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C86, C87, C88, C89, C90, C91, C92, C93, C94, C95, C96, C97, C98, C99, C100, C101, C102, C103, C104, C105, C106, C107, C108, C109, C110, C111, C112, C113, C114, C115, C116, C117, C118, C119, C120, C121, C122, C123, C124, C125, C126, C127, C128, C129, C130, C131, C132, C133, C134, C135, C136, C137, C138, C139, C140, C141, C142, C143, C144, C145, C146, C147, C148, C149, C150, C151, C152, C153, C154, C155, C156, C157, C158, C159, C160, C161, C162, C163, C164, C165, C166, C167, C168, C169, C170, C171, C172, C173, C174, C175, C176, C177, C178, C179, C180, C181, C182, C183, C184, C185, C186, C187, C188, C189, C190, C191, C192, C193, C194, C195, C196, C197, C198, C199, C200, C201, C202, C203, C204, C205, C206, C207, C208, C209, C210, C211, C212, C213, C214, C215, C216, C217, C218, C219, C220, C221, C222, C223, C224, C225, C226, C227, C228, C229, C230, C231, C232, C233, C234, C235, C236, C237, C238, C239, C240, C241, C242, C243, C244, C245, C246, C247, C248, C249, C250, C251, C252, C253, C254, C255, C256, C257, C258, C259, C260, C261, C262, C263, C264, C265, C266, C267, C268, C269, C270, C271, C272, C273, C274, C275, C276, C277, C278, C279, C280, C281, C282, C283, C284, C285, C286, C287, C288, C289, C290, C291, C292, C293, C294, C295, C296, C297, C298, C299, C300, C301, C302, C303, C304, C305, C306, C307, C308, C309, C310, C311, C312, C313, C314, C315, C316, C317, C318, C319, C320, C321, C322, C323, C324, C325, C326, C327, C328, C329, C330, C331, C332, C333, C334, C335, C336, C337, C338, C339, C340, C341, C342, C343, C344, C345, C346, C347, C348, C349, C350, C351, C352, C353, C354, C355, C356, C357, C358, C359, C360, C361, C362, C363, C364, C365, C366, C367, C368, C369, C370, C371, C372, C373, C374, C375, C376, C377, C378, C379, C380, C381, C382, C383, C384, C385, C386, C387, C388, C389, C390, C391, C392, C393, C394, C395, C396, C397, C398, C399, C400, C401, C402, C403, C404, C405, C406, C407, C408, C409, C410, C411, C412, C413, C414, C415, C416, C417, C418, C419, C420, C421, C422, C423, C424, C425, C426, C427, C428, C429, C430, C431, C432, C433, C434, C435, C436, C437, C438, C439, C440, C441, C442, C443, C444, C445, C446, C447, C448, C449, C450, C451, C452, C453, C454, C455, C456, C457, C458, C459, C460, C461, C462, C463, C464, C465, C466, C467, C468, C469, C470, C471, C472, C473, C474, C475, C476, C477, C478, C479, C480, C481, C482, C483, C484, C485, C486, C487, C488, C489, C490, C491, C492, C493, C494, C495, C496, C497, C498, C499, C500, C501, C502, C503, C504, C505, C506, C507, C508, C509, C510, C511, C512, C513, C514, C515, C516, C517, C518, C519, C520, C521, C522, C523, C524, C525, C526, C527, C528, C529, C530, C531, C532, C533, C534, C535, C536, C537, C538, C539, C540, C541, C542, C543, C544, C545, C546, C547, C548, C549, C550, C551, C552, C553, C554, C555, C556, C557, C558, C559, C560, C561, C562, C563, C564, C565, C566, C567, C568, C569, C570, C571, C572, C573, C574, C575, C576, C577, C578, C579, C580, C581, C582, C583, C584, C585, C586, C587, C588, C589, C590, C591, C592, C593, C594, C595, C596, C597, C598, C599, C600, C601, C602, C603, C604, C605, C606, C607, C608, C609, C610, C611, C612, C613, C614, C615, C616, C617, C618, C619, C620, C621, C622, C623, C624, C625, C626, C627, C628, C629, C630, C631, C632, C633, C634, C635, C636, C637, C638, C639, C640, C641, C642, C643, C644, C645, C646, C647, C648, C649, C650, C651, C652, C653, C654, C655, C656, C657, C658, C659, C660, C661, C662, C663, C664, C665, C666, C667, C668, C669, C670, C671, C672, C673, C674, C675, C676, C677, C678, C679, C680, C681, C682, C683, C684, C685, C686, C687, C688, C689, C690, C691, C692, C693, C694, C695, C696, C697, C698, C699, C700, C701, C702, C703, C704, C705, C

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/B00T or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull-downs to GND (PROJECT DESIGN CHOICE); reset mode M1/M0=00. AUX is input-only, no pull-down.

OLED: GPIO21/SDA and GPIO22/SCL are signal-only reservations; OLED VCC and I2C pull-ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 -> SN74AHCT1G125DBVR (OE=GND, V5) -> WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE. TPD45311 CC protection is not instantiated: its mandatory 2.7-4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test-point footprints/locations are deferred.

The schematic shows a PCB layout for a modular carrier. It includes various components like connectors (J1, J2, J3), integrated circuits (U1, U2, U3), capacitors (C1-C6), resistors (R8, R9), and test points (TP1-TP7). The layout is organized into sections labeled A, B, C, and D. Key features include DevKit USB-C ports (J1, J2), a universal E220-T22D socket (J3), and various power and signal management components. A title block at the bottom right identifies the project as "ESP32 E220 LoRa Receiver – Stage 5 modular carrier".

Sheet: /
File: esp32-e220.kicad_sch
Title: ESP32 E220 LoRa Receiver – Stage 5 modular carrier

Size:	A4	Date:	Rev:
KiCad E.D.A.	10.0.5		Id: 1/1

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

OLED: GPIO21/SDA and GPIO22/SCL are signal–only reservations; OLED VCC and I2C pull–ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 → SN74AHCT1G125DBVR (OE=GND, V5) → WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE; TPD45311 CC protection is not instantiated: its mandatory 2.7–4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test–point footprints/locations are deferred.

Sheet: /
File: esp32-e220.kicad_sch
Title: ESP32 E220 LoRa Receiver – Stage 5 modular carrier
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.5 Id: 1/1

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

OLED: GPIO21/SDA and GPIO22/SCL are signal–only reservations; OLED VCC and I2C pull–ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 → SN74AHCT1G125DBVR (OE=GND, V5) → WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE; TPD45311 CC protection is not instantiated: its mandatory 2.7–4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test–point footprints/locations are deferred.

Sheet: /
File: esp32-e220.kicad_sch
Title: ESP32 E220 LoRa Receiver – Stage 5 modular carrier
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.5 Id: 1/1

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

OLED: GPIO21/SDA and GPIO22/SCL are signal–only reservations; OLED VCC and I2C pull–ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 → SN74AHCT1G125DBVR (OE=GND, V5) → WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE; TPD45311 CC protection is not instantiated: its mandatory 2.7–4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test–point footprints/locations are deferred.

Sheet: /
File: esp32-e220.kicad_sch
Title: ESP32 E220 LoRa Receiver – Stage 5 modular carrier
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.5 Id: 1/1

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

OLED: GPIO21/SDA and GPIO22/SCL are signal–only reservations; OLED VCC and I2C pull–ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 → SN74AHCT1G125DBVR (OE=GND, V5) → WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE; TPD45311 CC protection is not instantiated: its mandatory 2.7–4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test–point footprints/locations are deferred.

Sheet: /
File: esp32-e220.kicad_sch
Title: ESP32 E220 LoRa Receiver – Stage 5 modular carrier
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.5 Id: 1/1

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

OLED: GPIO21/SDA and GPIO22/SCL are signal–only reservations; OLED VCC and I2C pull–ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 → SN74AHCT1G125DBVR (OE=GND, V5) → WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE; TPD45311 CC protection is not instantiated: its mandatory 2.7–4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test–point footprints/locations are deferred.

Sheet: /
File: esp32-e220.kicad_sch
Title: ESP32 E220 LoRa Receiver – Stage 5 modular carrier
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.5 Id: 1/1

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

OLED: GPIO21/SDA and GPIO22/SCL are signal–only reservations; OLED VCC and I2C pull–ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 → SN74AHCT1G125DBVR (OE=GND, V5) → WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE; TPD45311 CC protection is not instantiated: its mandatory 2.7–4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test–point footprints/locations are deferred.

Sheet: /
File: esp32-e220.kicad_sch
Title: ESP32 E220 LoRa Receiver – Stage 5 modular carrier
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.5 Id: 1/1

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

OLED: GPIO21/SDA and GPIO22/SCL are signal–only reservations; OLED VCC and I2C pull–ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 → SN74AHCT1G125DBVR (OE=GND, V5) → WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE; TPD45311 CC protection is not instantiated: its mandatory 2.7–4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test–point footprints/locations are deferred.

Sheet: /
File: esp32-e220.kicad_sch
Title: ESP32 E220 LoRa Receiver – Stage 5 modular carrier
Size: A4 Date: Rev:
KiCad E.D.A. 10.0.5 Id: 1/1

Stage 5 – main board only. DevKit USB–C programming and main–board USB–C power are mutually exclusive (approved Rev A policy).

No bare ESP32, 3V3 buck, EN/BOOT or programming circuit on this PCB. J1/J2 are the verified removable DevKit sockets.

J3 is universal E220–T22D interface: fit one E220–400T22D OR E220–900T22D. No socket/footprint is selected.

E220 M0/M1 use 10k pull–downs to GND (PROJECT DESIGN CHOICE): reset mode M1/M0=00. AUX is input–only, no pull–down.

OLED: GPIO21/SDA and GPIO22/SCL are signal–only reservations; OLED VCC and I2C pull–ups are NC/DNP in Rev A.

TUSB320 UFP/GPIO: PORT=GND, ADDR=NC, EN_N=GND. OUT1/Q1 gates TPS259630: Default/detach OFF; Medium/High ON.

USB4105–GF–A: all A/B VBUS, GND and shell contacts are explicit; D+/D– are local NC nets, SBU pins NC, no ESP USB connection.

GPIO4 → SN74AHCT1G125DBVR (OE=GND, V5) → WS2812B–V5 DIN. No WS2812 footprint assigned at schematic stage.

D3 TPD1E10B06DPYR protects VBUS_PRE; TPD45311 CC protection is not instantiated: its mandatory 2.7–4.5V VPWR rail is unresolved in this gate architecture.

TP1 5V_SYS; TP2 E220_VCC(5V_SYS); TP3 E220_TXD; TP4 M1; TP5 AUX; TP6 E220_RXD; TP7 E220_TXD. Test–point footprints/locations are deferred.